

MATH 162, Winter 2026
SCRIPT 12: Derivatives

Throughout this sheet, we let $f: A \rightarrow \mathbb{R}$ be a real valued function with domain $A \subset \mathbb{R}$. We also now assume the domain $A \subset \mathbb{R}$ is open.

Definition 12.1. The *derivative* of f at a point $a \in A$ is the number $f'(a)$ defined by the following limit:

$$f'(a) = \lim_{h \rightarrow 0} \frac{f(a+h) - f(a)}{h},$$

provided the limit on the right hand side exists. If $f'(a)$ exists, we say that f is *differentiable at* a . If f is differentiable at all points of its domain, we say that f is *differentiable*. In this case, the values $f'(a)$ define a new function $f': A \rightarrow \mathbb{R}$ called the *derivative* of f .

Remark 12.2. If A is not open the limit in Definition 12.1 may not exist. For example, if $f: [a, b] \rightarrow \mathbb{R}$ then we cannot define the derivative at the endpoints. For any c in the domain of f , we define the *right-hand-derivative* $f'_+(c)$ and the *left-hand-derivative* $f'_-(c)$ by

$$f'_+(c) = \lim_{h \rightarrow 0^+} \frac{f(c+h) - f(c)}{h} \quad \text{and} \quad f'_-(c) = \lim_{h \rightarrow 0^-} \frac{f(c+h) - f(c)}{h}.$$

We say that f is differentiable on $[a, b]$ if f is differentiable on (a, b) and $f'_+(a)$ and $f'_-(b)$ exist.

Lemma 12.3. Let $a \in \mathbb{R}$. Then

$$\lim_{x \rightarrow a} f(x) = \lim_{h \rightarrow 0} f(a+h),$$

assuming that one of the two limits exists. (So if the limit on the left exists, so does the one on the right, and they are equal. Similarly, if the limit on the right exists, then so does the one on the left, and they are equal.)

Proof of Lemma 12.3. Let $a \in \mathbb{R}$. We want to show $\lim_{x \rightarrow a} f(x) = \lim_{h \rightarrow 0} f(a+h)$.

($\Rightarrow$) Let $\lim_{x \rightarrow a} f(x)$ exist. Then, we have that $\forall \epsilon > 0, \exists \delta > 0$ such that if $x \in A$ and $0 < |x - a| < \delta$ then $|f(x) - L| < \epsilon$.

Now, let $x = h + a$ where $x, h, a \in \mathbb{R}$. Then, as $\lim_{x \rightarrow a} f(x)$ exists and $x = h + a$, we have that $\forall \epsilon > 0, \exists \delta > 0$ where $h+a \in \mathbb{R}$. Then, by definition, we have that $h+a \in \mathbb{R}$ and $0 < |(h+a) - a| < \delta$. We can rewrite $0 < |(h+a) - a| < \delta$ to get $0 < |h - 0| < \delta$. Now, as we have that $\lim_{x \rightarrow a} f(x)$ exist, we have that $|f(x) - L| < \epsilon$.

As we have merely found an equivalent variable to x , the limit still exists, and the conditions stay true. Therefore, we have that $\forall \epsilon > 0, \exists \delta > 0$ such that if $h+a \in \mathbb{R}$ and $0 < |(h+a) - a| = |h - 0| < \delta$, then $|f(x) - L| < \epsilon$. Thus, we have that $\lim_{h \rightarrow 0} f(a+h)$.

($\Leftarrow$) Let $\lim_{h \rightarrow 0} f(a+h)$ exists. Then, we have that $\forall \epsilon > 0, \exists \delta > 0$ such that if $a+h \in A$ and $0 < |h-0| < \delta$ then $|f(x) - L| < \epsilon$.

Let $h = x - a$ where $x, h, a \in \mathbb{R}$. Then, as $\lim_{h \rightarrow 0} f(a+h)$ exists and $h = x - a$, we have that $\forall \epsilon > 0, \exists \delta > 0$ where $x \in \mathbb{R}$. Then, by definition, we have that $z \in \mathbb{R}$ and $0 < |x - a| < \delta$. Now, as we have that $\lim_{h \rightarrow 0} f(a+h)$ exist, we have that $|f(x) - L| < \epsilon$. Then, as $h = x - a$ we can simply substitute h with $x - a$ to get $\forall \epsilon > 0, \exists \delta > 0$ where if $x \in A$ and $0 < |x - a| < \delta$ then $|f(x) - L| < \epsilon$. This is the same as saying $\lim_{x \rightarrow a} f(x)$ exists.

This suffices to prove that $\lim_{x \rightarrow a} f(x) = \lim_{h \rightarrow 0} f(a+h)$. $\square$

Here is a useful reformulation of the definition of the derivative.

Theorem 12.4. *Let $a \in \mathbb{R}$. Then f is differentiable at a if and only if $\lim_{x \rightarrow a} \frac{f(x) - f(a)}{x - a}$ exists. Moreover, if f is differentiable at a , then the derivative of f at a is given by this limit:*

$$f'(a) = \lim_{x \rightarrow a} \frac{f(x) - f(a)}{x - a}.$$

Proof of 12.4. Let $a \in \mathbb{R}$. We want to show f is differentiable at a if and only if $\lim_{x \rightarrow a} \frac{f(x) - f(a)}{x - a}$.

($\Leftarrow$) Let $\frac{f(x) - f(a)}{x - a}$ exist. Then, let $x = h + a$. By **Lemma 12.3**, we have that $\lim_{x \rightarrow a} f(x) = \lim_{h \rightarrow 0} f(a+h)$. Therefore, $\lim_{x \rightarrow a} \frac{f(x) - f(a)}{x - a} = \lim_{h \rightarrow 0} \frac{f(a+h) - f(a)}{h}$ (as $x - a = h$). Thus, by **Definition 12.1**, we have that $f'(a) = \lim_{h \rightarrow 0} \frac{f(a+h) - f(a)}{h}$.

($\Rightarrow$) Let f be differentiable at a . Then, we have that $f'(a)$ exists. Then, we have that the limit $\lim_{h \rightarrow 0} \frac{f(a+h) - f(a)}{h}$ is present. Then, by **Lemma 12.3**, we have that $\lim_{h \rightarrow 0} f(a+h) = \lim_{x \rightarrow a} f(x)$. Let $h = x - a$. Then, we can write $\lim_{h \rightarrow 0} \frac{f(a+h) - f(a)}{h}$ as $\lim_{x \rightarrow a} \frac{f(x) - f(a)}{x - a}$. Thus, we get that the derivative of f at a is given by $\lim_{x \rightarrow a} \frac{f(x) - f(a)}{x - a}$. $\square$

Theorem 12.5. *If f is differentiable at a , then f is continuous at a .*

Proof of Theorem 12.5. We want to show that if f is differentiable at a then f is continuous at a .

Let f be differentiable at a . Then, we have that $\lim_{h \rightarrow 0} \frac{f(a+h)-f(a)}{h} = f'(a)$ exists. We can therefore write

$$f(a+h) - f(a) = \frac{f(a+h) - f(a)}{h} \cdot h.$$

Then, we may take the limits as $h \rightarrow 0$, and, by **Theorem 11.9** we get that

$$\lim_{h \rightarrow 0} [f(a+h) - f(a)] = \lim_{h \rightarrow 0} \frac{f(a+h) - f(a)}{h} \cdot \lim_{h \rightarrow 0} h = f'(a) \cdot 0 = 0.$$

Thus, we get that $\lim_{h \rightarrow 0} f(a+h) = f(a)$. This suffices to prove that f is continuous at a . $\square$

Exercise 12.6. Show that the converse of Theorem 12.5 is not true.

Exercise 12.6. We want to show that if f is continuous at a , it is not necessarily the case that f is differentiable at a . We do this by counterexample.

Consider $f(x) = |x|$ where $f : \mathbb{R} \rightarrow \mathbb{R}$, and let $a = 0$. We want to show that f is continuous at 0 but not differentiable at 0.

Suppose f is continuous at 0. Then, we have that $\forall \epsilon > 0, \exists \delta > 0$ such that if $x \in \mathbb{R}$ and $0 < |x - a| < \delta$ then $0 < |f(x) - f(a)| < \epsilon$. Let $\delta = \epsilon$. Then, we have that $|f(x) - f(0)| = |x| < \epsilon$ and that $|x| < \delta$.

Then, we want to show that f is not differentiable at 0. By **Theorem 12.4** it suffices to show that the limit

$$\lim_{x \rightarrow 0} \frac{f(x) - f(0)}{x - 0} = \lim_{x \rightarrow 0} \frac{|x|}{x}$$

exists. For this, we can compute the one-sided limits. For $x > 0$, $|x| = x$, so

$$\lim_{x \rightarrow 0^+} \frac{|x|}{x} = 1.$$

For $x < 0$, $|x| = -x$, so

$$\lim_{x \rightarrow 0^-} \frac{|x|}{x} = -1.$$

As we do $1 \neq -1$, the limit does not exist, and therefore f is not differentiable at 0. $\square$

Exercise 12.7. Show that, for all $n \in \mathbb{N}$,

$$x^n - a^n = (x - a)(x^{n-1} + ax^{n-2} + a^2x^{n-3} + \cdots + a^{n-2}x + a^{n-1}),$$

or equivalently,

$$x^n - a^n = (x - a) \left(\sum_{i=0}^{n-1} x^{n-1-i} a^i \right).$$

Exercise 12.7. We want to show that $\forall n \in \mathbb{N}$,

$$x^n - a^n = (x - a) \left(\sum_{i=0}^{n-1} x^{n-1-i} a^i \right).$$

Consider the expression $(x - a) \left(\sum_{i=0}^{n-1} x^{n-1-i} a^i \right)$. We can expand this to get

$$(x - a)(x^{n-1}) + (x - a)(ax^{n-2}) + (x - a)(a^2x^{n-3}) + \dots + (x - a)(a^{n-2}x) + (x - a)(a^{n-1}).$$

We can further expand each term to get

$$(x^n - ax^{n-1}) + (ax^{n-1} - a^2x^{n-2}) + (a^2x^{n-2} - a^3x^{n-3}) + \dots + (a^{n-2}x^2 - a^{n-1}x) + (a^{n-1}x - a^n).$$

When simplifying this expression, we get that all intermediate terms cancel out: $-ax^{n-1}$ cancels with ax^{n-1} , $-a^2x^{n-2}$ cancels with a^2x^{n-2} , and so on until $-a^{n-1}x$ cancels with $a^{n-1}x$. Therefore, we are left with $x^n - a^n$, which completes the proof. $\square$

Exercise 12.8. a) Let $n \in \mathbb{N}$. Suppose $f : \mathbb{R} \rightarrow \mathbb{R}$ is given by $f(x) = x^n$. Use Exercise 12.7 to prove that $f'(a) = na^{n-1}$, for all $a \in \mathbb{R}$.

b) Let $k \in \mathbb{R}$. Prove that if $f : \mathbb{R} \rightarrow \mathbb{R}$ is given by $f(x) = k$, then $f'(a) = 0$, for all $a \in \mathbb{R}$.

Proof. Exercise 12.8(a) Let $n \in \mathbb{N}$. Suppose $f : \mathbb{R} \rightarrow \mathbb{R}$ is given by $f(x) = x^n$. We want prove that $f'(a) = na^{n-1}$, for all $a \in \mathbb{R}$.

As $f(x) = x^n$, and we have f is differentiable (at a ?). By **Theorem 12.4**, we can write $f'(a)$ as $\lim_{x \rightarrow a} \frac{x^n - a^n}{x - a}$. By **Exercise 12.7**, we have that $x^n - a^n = (x - a)(x^{n-1} + ax^{n-2} + a^2x^{n-3} + \dots + a^{n-2}x + a^{n-1})$. Therefore, we have that $\lim_{x \rightarrow a} \frac{x^n - a^n}{x - a} = \lim_{x \rightarrow a} \frac{(x - a)(x^{n-1} + ax^{n-2} + a^2x^{n-3} + \dots + a^{n-2}x + a^{n-1})}{(x - a)}$

We can simplify this expression to $\lim_{x \rightarrow a} (x^{n-1} + ax^{n-2} + a^2x^{n-3} + \dots + a^{n-2}x + a^{n-1})$. Then, we have that $\lim_{x \rightarrow a} \left(\sum_{i=0}^{n-1} x^{n-1-i} a^i \right)$. Then, we can write this as $(\sum_{i=0}^{n-1} a^{n-1-i}) = na^{n-1}$. $\square$

Exercise 12.8(b). Let $k \in \mathbb{R}$. We want to if $f : \mathbb{R} \rightarrow \mathbb{R}$ is given by $f(x) = k$, then $f'(a) = 0$, $\forall a \in \mathbb{R}$.

Let $f(x) = k$. We know $f'(a) = \lim_{x \rightarrow a} \frac{f(x) - f(a)}{x - a}$. Then, we can substitute $f(x) = k$ and $f(a) = k$ to get $\lim_{x \rightarrow a} \frac{k - k}{x - a} = \lim_{x \rightarrow a} \frac{0}{x - a} = 0$. Therefore, we $f'(a) = 0$. $\square$

Exercise 12.9. Suppose that $f: A \rightarrow \mathbb{R}$ and $g: A \rightarrow \mathbb{R}$ are differentiable at $a \in A$.

- (a) Prove that $(f + g)$ is differentiable at a and compute $(f + g)'(a)$ in terms of $f'(a)$ and $g'(a)$.
- (b) Prove that (fg) is differentiable at a and compute $(fg)'(a)$ in terms of $f(a)$, $g(a)$, $f'(a)$ and $g'(a)$.
- (c) Prove that $\frac{1}{g}$ is differentiable at a (under an appropriate assumption) and compute $(\frac{1}{g})'(a)$ in terms of $g'(a)$ and $g(a)$. What assumption do you need to make?
- (d) Prove that $\frac{f}{g}$ is differentiable at a (under an appropriate assumption) and compute $(\frac{f}{g})'(a)$ in terms of $f(a)$, $g(a)$, $f'(a)$ and $g'(a)$. What assumption do you need to make?

Exercise 12.9(a.) Let $f: A \rightarrow \mathbb{R}$ and $g: A \rightarrow \mathbb{R}$ are differentiable at $a \in A$. We want to show $(f + g)$ is differentiable at a .

Consider $(f + g)(x)$. To show that $(f + g)(x)$ is differentiable at a , it suffices to show that

$$(f + g)'(a) = \lim_{x \rightarrow a} \frac{(f + g)(x) - (f + g)(a)}{x - a}$$

As $(f + g)$ can be written as $f(x) + g(x)$, we can expand the equation to get

$$(f + g)'(a) = \lim_{x \rightarrow a} \frac{f(x) + g(x) - f(a) - g(a)}{x - a} = \lim_{x \rightarrow a} \frac{f(x) - f(a)}{x - a} + \lim_{x \rightarrow a} \frac{g(x) - g(a)}{x - a}$$

As both f and g are differentiable, we get that this equation can be simplified to get

$$(f + g)'(a) = f'(a) + g'(a)$$

where $f'(a), g'(a) \in \mathbb{R}$.

Therefore, we get that ... [is this enough]

□

Exercise 12.9(b.) Let $f: A \rightarrow \mathbb{R}$ and $g: A \rightarrow \mathbb{R}$ are differentiable at $a \in A$. We want to show (fg) is differentiable at a .

Consider $(fg)(x)$. To show that $(fg)(x)$ is differentiable at a , it suffices to show that

$$(fg)'(a) = \lim_{x \rightarrow a} \frac{(fg)(x) - (fg)(a)}{x - a}$$

We can expand this equation to get

$$(fg)'(a) = \lim_{x \rightarrow a} \frac{(fg)(x) - (fg)(a)}{x - a} = \lim_{x \rightarrow a} \frac{f(x) \cdot g(x) - f(x)g(a) - f(a) \cdot g(a) + f(x)g(a)}{x - a}$$

We can further break this up to get

$$fg'(a) = \lim_{x \rightarrow a} \frac{f(x)[g(x) - g(a)]}{x - a} + \lim_{x \rightarrow a} \frac{g(a)[f(x) - f(a)]}{x - a}$$

Now, we can simplify this to get

$$fg'(a) = \lim_{x \rightarrow a} f(x) \cdot \lim_{x \rightarrow a} \frac{g(x) - g(a)}{x - a} + \lim_{x \rightarrow a} g(a) \cdot \lim_{x \rightarrow a} \frac{f(x) - f(a)}{x - a} = f(a)g'(a) + g(a)f'(a)$$

Hence, we have shown that the limit $\lim_{x \rightarrow a} \frac{(fg)(x) - (fg)(a)}{x - a}$ exists and equals $f(a)g'(a) + g(a)f'(a)$. $\square$

Exercise 12.9(c). Let $g: A \rightarrow \mathbb{R}$ are differentiable at $a \in A$. We want to show $\frac{1}{g}$ is differentiable at a .

Consider $\frac{1}{g}(x)$. To show that $\frac{1}{g}(x)$ is differentiable at a , it suffices to show that:

$$\frac{1}{g}'(x) = \lim_{x \rightarrow a} \frac{\frac{1}{g}(x) - \frac{1}{g}(a)}{x - a}$$

We can rewrite the expression $\frac{\frac{1}{g}(x) - \frac{1}{g}(a)}{x - a}$ to get

$$\frac{1}{g}'(x) = \lim_{x \rightarrow a} \frac{\frac{1}{g}(x) - \frac{1}{g}(a)}{x - a} = \lim_{x \rightarrow a} \frac{g(x) - g(a)}{(x - a) \cdot g(x) \cdot g(a)}$$

Furthermore, we get further expand this to get

$$\frac{1}{g}'(x) = \lim_{x \rightarrow a} -\frac{1}{g(x)g(a)} \cdot \frac{g(x) - g(a)}{x - a}$$

Then, we can solve the limit to get

$$\frac{1}{g}'(x) = -\frac{1}{(g(a))^2} \cdot g'(a)$$

As we have show that the limit exists, it suffices to prove that $\frac{1}{g}$ is differentiable at a . However, for this, we must assume that $g(a) \neq 0$. $\square$

Exercise 12.9(d). Let $f: A \rightarrow \mathbb{R}$ and $g: A \rightarrow \mathbb{R}$ be differentiable at $a \in A$. We want to show $\frac{f}{g}$ is differentiable at a .

Consider $\frac{f}{g}(x)$. To show that $\frac{f}{g}(x)$ is differentiable at a , it suffices to show that:

$$\left(\frac{f}{g}\right)'(a) = \lim_{x \rightarrow a} \frac{\frac{f}{g}(x) - \frac{f}{g}(a)}{x - a}$$

We can rewrite $\frac{f}{g} = f \cdot \frac{1}{g}$, and apply the product rule from Exercise 12.9(b) to get

$$\left(\frac{f}{g}\right)'(a) = \left(f \cdot \frac{1}{g}\right)'(a) = f'(a) \cdot \frac{1}{g}(a) + f(a) \cdot \left(\frac{1}{g}\right)'(a)$$

Then, applying the result of Exercise 12.9(c), we can further expand this to get

$$\left(\frac{f}{g}\right)'(a) = \frac{f'(a)}{g(a)} - \frac{f(a)g'(a)}{(g(a))^2}$$

We can simplify this to get

$$\left(\frac{f}{g}\right)'(a) = \frac{f'(a)g(a) - f(a)g'(a)}{(g(a))^2}$$

As we have shown that the limit exists, it suffices to prove that $\frac{f}{g}$ is differentiable at a . However, for this, we must assume that $g(a) \neq 0$. □

One of the most important results concerning the differentiation of functions is the rule for the derivative of a composition of functions. Let $f: B \rightarrow \mathbb{R}$, $g: A \rightarrow \mathbb{R}$ be functions such that $g(A) \subset B$. The composition $(f \circ g)(x) = f(g(x))$ is defined for all $x \in A$.

Theorem 12.10. *Let $a \in A$, $g: A \rightarrow \mathbb{R}$ and $f: I \rightarrow \mathbb{R}$, where I is an interval containing $g(A)$. Suppose that g is differentiable at a and f is differentiable at $g(a)$. Then $f \circ g$ is differentiable at a and:*

$$(f \circ g)'(a) = f'(g(a)) \cdot g'(a).$$

Hint: The following function could be useful: $\varphi: I \rightarrow \mathbb{R}$ defined by

$$\varphi(y) = \begin{cases} \frac{f(y) - f(g(a))}{y - g(a)} & \text{if } y \neq g(a), \\ f'(g(a)) & \text{if } y = g(a). \end{cases}$$

Proof of Theorem 12.10. We want to show that $f \circ g$ is differentiable at a and that $(f \circ g)'(a) = f'(g(a)) \cdot g'(a)$.

Consider the auxiliary function $\varphi: I \rightarrow \mathbb{R}$ defined by

$$\varphi(y) = \begin{cases} \frac{f(y) - f(g(a))}{y - g(a)} & \text{if } y \neq g(a), \\ f'(g(a)) & \text{if } y = g(a). \end{cases}$$

We first show that φ is continuous at $g(a)$. By differentiability of f at $g(a)$, we have that

$$\lim_{y \rightarrow g(a)} \frac{f(y) - f(g(a))}{y - g(a)} = f'(g(a)) = \varphi(g(a)),$$

hence φ is continuous at $g(a)$.

Now, we can write the following identity for all $x \neq a$:

$$\frac{f(g(x)) - f(g(a))}{x - a} = \varphi(g(x)) \cdot \frac{g(x) - g(a)}{x - a}.$$

Indeed, if $g(x) \neq g(a)$, the right hand side becomes $\frac{f(g(x)) - f(g(a))}{g(x) - g(a)} \cdot \frac{g(x) - g(a)}{x - a}$, which equals the left hand side by cancellation. If $g(x) = g(a)$, both sides equal 0.

Taking limits as $x \rightarrow a$, we have that

$$(f \circ g)'(a) = \lim_{x \rightarrow a} \frac{f(g(x)) - f(g(a))}{x - a} = \lim_{x \rightarrow a} \varphi(g(x)) \cdot \lim_{x \rightarrow a} \frac{g(x) - g(a)}{x - a}.$$

The second limit equals $g'(a)$ by differentiability of g at a . For the first limit, we have that g is differentiable at a , hence continuous at a by Theorem 12.5, and therefore $g(x) \rightarrow g(a)$ as $x \rightarrow a$. Since φ is continuous at $g(a)$, we have that $\lim_{x \rightarrow a} \varphi(g(x)) = \varphi(g(a)) = f'(g(a))$.

Therefore, $(f \circ g)'(a) = f'(g(a)) \cdot g'(a)$. This suffices to complete the proof. $\square$

We now come to the most important theorem in differential calculus, Corollary 12.15.

Definition 12.11. Let $f: A \rightarrow \mathbb{R}$ be a function. If $f(a)$ is the last point of $f(A)$, then $f(a)$ is called the *maximum value* of f . If $f(a)$ is the first point of $f(A)$, then $f(a)$ is the *minimum value* of f . We say that $f(a)$ is a *local maximum value* of f if there exists a region R containing a such that $f(a)$ is the last point of $f(A \cap R)$. We say that $f(a)$ is a *local minimum value* of f if there exists a region R containing a such that $f(a)$ is the first point of $f(A \cap R)$.

Remark 12.12. Equivalently, $f(a)$ is a *local maximum* (resp. *minimum*) value of f if there exists U open in A such that $f(a)$ is the last (resp. first) point of $f(U)$.

Theorem 12.13. Let $f: A \rightarrow \mathbb{R}$ be differentiable at a . Suppose that $f(a)$ is a local maximum or local minimum value of f . Then $f'(a) = 0$.

Proof of Theorem 12.13. Let $f: A \rightarrow \mathbb{R}$ be differentiable at a , and suppose that $f(a)$ is a local maximum value of f . Then, there exists a region R containing a such that for all $x \in A \cap R$, we have $f(x) \leq f(a)$.

Now, let $x \in A$ with $x \neq a$. We consider two cases.

If $x > a$, then $x - a > 0$ and $f(x) - f(a) \leq 0$, hence we have that

$$\frac{f(x) - f(a)}{x - a} \leq 0.$$

Taking limits as $x \rightarrow a^+$, and using differentiability of f at a , we have that $f'(a) \leq 0$.

If $x < a$, then $x - a < 0$ and $f(x) - f(a) \leq 0$, hence we have that

$$\frac{f(x) - f(a)}{x - a} \geq 0.$$

Taking limits as $x \rightarrow a^-$, and using differentiability of f at a , we have that $f'(a) \geq 0$.

Therefore, we have that $f'(a) \leq 0$ and $f'(a) \geq 0$, which gives us $f'(a) = 0$. The case where $f(a)$ is a local minimum is analogous. This suffices to complete the proof. $\square$

Theorem 12.14. *Suppose that $f: [a, b] \rightarrow \mathbb{R}$ is continuous, differentiable on (a, b) and that $f(a) = f(b)$. Then there exists a point $\lambda \in (a, b)$ such that $f'(\lambda) = 0$.*

Proof of theorem 12.14. Let $f: [a, b] \rightarrow \mathbb{R}$ be continuous on $[a, b]$ and differentiable on (a, b) , with $f(a) = f(b)$.

Since f is continuous on a closed interval, it attains a maximum and minimum on $[a, b]$. Let $x_{\max}, x_{\min} \in [a, b]$ be such that

$$f(x_{\max}) \geq f(x) \geq f(x_{\min}) \quad \text{for all } x \in [a, b].$$

If f is constant, then $f'(x) = 0$ for all $x \in (a, b)$, and we are done. Otherwise, at least one of $x_{\max}$ or $x_{\min}$ lies in (a, b) . At that point, f has a local maximum or minimum. By **Theorem 12.13**, we have

$$f'(\lambda) = 0$$

for some $\lambda \in (a, b)$. $\square$

Corollary 12.15. *Suppose that $f: [a, b] \rightarrow \mathbb{R}$ is continuous on $[a, b]$ and differentiable on (a, b) . Then there exists a point $\lambda \in (a, b)$ such that:*

$$f(b) - f(a) = f'(\lambda)(b - a).$$

Proof of Corollary 12.15. Suppose that $f: [a, b] \rightarrow \mathbb{R}$ is continuous on $[a, b]$ and differentiable on (a, b) . We want to show that there exists a point $\lambda \in (a, b)$ such that $f(b) - f(a) = f'(\lambda)(b - a)$.

Firstly, we can define a function $g: [a, b] \rightarrow \mathbb{R}$ by

$$g(x) = f(x) - f(a) - \left(\frac{f(b) - f(a)}{b - a} \right) (x - a).$$

Since f is continuous on $[a, b]$ and differentiable on (a, b) , and the remaining terms are linear, it follows that g is continuous on $[a, b]$ and differentiable on (a, b) .

Now, we evaluate g at the endpoints. First,

$$g(a) = f(a) - f(a) - \left(\frac{f(b) - f(a)}{b - a} \right) (a - a) = 0.$$

Similarly,

$$g(b) = f(b) - f(a) - \left(\frac{f(b) - f(a)}{b - a} \right) (b - a) = 0.$$

Thus, we have $g(a) = g(b)$.

Since g is continuous on $[a, b]$, differentiable on (a, b) , and satisfies $g(a) = g(b)$, we may apply **Theorem 12.14**. Hence, there exists some $\lambda \in (a, b)$ such that

$$g'(\lambda) = 0.$$

Now, differentiating g , we obtain

$$g'(x) = f'(x) - \frac{f(b) - f(a)}{b - a}.$$

Therefore, at $x = \lambda$, we have

$$0 = g'(\lambda) = f'(\lambda) - \frac{f(b) - f(a)}{b - a}.$$

Rearranging gives

$$f'(\lambda) = \frac{f(b) - f(a)}{b - a}.$$

Multiplying both sides by $(b - a)$, we obtain

$$f(b) - f(a) = f'(\lambda)(b - a),$$

This suffices to complete the proof. □

Corollary 12.16. *Suppose that $f: [a, b] \rightarrow \mathbb{R}$ is continuous on $[a, b]$ and differentiable on (a, b) . Then,*

- (a) *If $f'(x) > 0$ for all $x \in (a, b)$, then f is strictly increasing on $[a, b]$.*
- (b) *If $f'(x) < 0$ for all $x \in (a, b)$, then f is strictly decreasing on $[a, b]$.*
- (c) *If $f'(x) = 0$ for all $x \in (a, b)$, then f is constant on $[a, b]$.*

Proof of (a). Let $x \in (a, b]$. Since f is continuous on $[a, b]$, it is continuous on the sub-interval $[a, x]$. Since f is differentiable on (a, b) , it is differentiable on (a, x) . Thus by the Mean Value Theorem (Corollary 12.15), there exists a point $\lambda \in (a, x) \subset (a, b)$ such that

$$f(x) - f(a) = f'(\lambda)(x - a).$$

By hypothesis, $f'(\lambda) > 0$, and since $x > a$ we have $x - a > 0$. Therefore

$$f(x) - f(a) = f'(\lambda)(x - a) > 0,$$

so $f(x) > f(a)$. Since $x \in (a, b]$ was arbitrary, we have $f(x) > f(a)$ for all $x \in (a, b]$, and the case $x = a$ holds trivially. Hence f is strictly increasing on $[a, b]$. $\square$

Proof of (b). Let $x \in (a, b]$. By the same reasoning as part (a), f restricted to $[a, x]$ satisfies the hypotheses of the Mean Value Theorem (Corollary 12.15), so there exists $\lambda \in (a, x) \subset (a, b)$ such that

$$f(x) - f(a) = f'(\lambda)(x - a).$$

By hypothesis, $f'(\lambda) < 0$, and since $x - a > 0$, we have

$$f(x) - f(a) = f'(\lambda)(x - a) < 0,$$

so $f(x) < f(a)$. Since $x \in (a, b]$ was arbitrary, f is strictly decreasing on $[a, b]$. $\square$

Proof of (c). Let $x \in (a, b]$. By the same reasoning as part (a), there exists $\lambda \in (a, x) \subset (a, b)$ such that

$$f(x) - f(a) = f'(\lambda)(x - a).$$

By hypothesis, $f'(\lambda) = 0$, so

$$f(x) - f(a) = 0 \cdot (x - a) = 0,$$

and hence $f(x) = f(a)$. Since $x \in (a, b]$ was arbitrary, $f(x) = f(a)$ for all $x \in [a, b]$, so f is constant on $[a, b]$. $\square$

Remark 12.17. Corollary 12.16 also holds if, throughout the statement of the result, $[a, b]$ is replaced by an interval I , and (a, b) is replaced by the interior of I .

Corollary 12.18. Suppose that $f: [a, b] \rightarrow \mathbb{R}$ and $g: [a, b] \rightarrow \mathbb{R}$ are continuous on $[a, b]$, differentiable on (a, b) , and $f'(x) = g'(x)$ for all $x \in (a, b)$. Then there is some $c \in \mathbb{R}$ such that for all $x \in [a, b]$, we have $f(x) = g(x) + c$.

Proof of Corollary 12.18. We want to show that there exists some $c \in \mathbb{R}$ such that $f(x) = g(x) + c$ for all $x \in [a, b]$.

Let $h(x) = f(x) - g(x)$. Since f and g are differentiable on (a, b) , by **Exercise 12.9(a)** we have that $h'(x) = f'(x) - g'(x)$. Since $f'(x) = g'(x)$ for all $x \in (a, b)$, it follows that $h'(x) = 0$ for all $x \in (a, b)$. Therefore, by Corollary 12.16(c), h is constant on $[a, b]$, and so $h(x) = c$ for some $c \in \mathbb{R}$. Therefore $f(x) - g(x) = c$, and hence $f(x) = g(x) + c$ for all $x \in [a, b]$. $\square$

Corollary 12.19. *Suppose that $f: [a, b] \rightarrow \mathbb{R}$ and $g: [a, b] \rightarrow \mathbb{R}$ are continuous on $[a, b]$ and differentiable on (a, b) . Then there exists a point $\lambda \in (a, b)$ such that:*

$$(f(b) - f(a))g'(\lambda) = (g(b) - g(a))f'(\lambda).$$

Proof of Corollary 12.19. We want to show that there exists a point $\lambda \in (a, b)$ such that $(f(b) - f(a))g'(\lambda) = (g(b) - g(a))f'(\lambda)$.

We can define $h: [a, b] \rightarrow \mathbb{R}$ by

$$h(x) = (f(b) - f(a))g(x) - (g(b) - g(a))f(x).$$

Since f and g are continuous on $[a, b]$ and differentiable on (a, b) , it follows that h is continuous on $[a, b]$ and differentiable on (a, b) .

Now, we want to show that $h(a) = h(b)$. Computing directly, we get

$$\begin{aligned} h(a) &= (f(b) - f(a))g(a) - (g(b) - g(a))f(a) \\ &= f(b)g(a) - f(a)g(a) - g(b)f(a) + g(a)f(a) \\ &= f(b)g(a) - g(b)f(a), \end{aligned}$$

and similarly,

$$\begin{aligned} h(b) &= (f(b) - f(a))g(b) - (g(b) - g(a))f(b) \\ &= f(b)g(b) - f(a)g(b) - g(b)f(b) + g(a)f(b) \\ &= g(a)f(b) - f(a)g(b). \end{aligned}$$

Therefore $h(a) = h(b)$. Since h is continuous on $[a, b]$, differentiable on (a, b) , and satisfies $h(a) = h(b)$, we may apply **Theorem 12.14**. Hence, there exists $\lambda \in (a, b)$ such that $h'(\lambda) = 0$, which is the desired conclusion. $\square$

Finally, we prove another very important theorem that tells us about inverse functions and their derivatives.

Theorem 12.20. *Suppose that $f: (a, b) \rightarrow \mathbb{R}$ is differentiable and that the derivative $f': (a, b) \rightarrow \mathbb{R}$ is continuous. Also suppose that there is a point $p \in (a, b)$ such that $f'(p) \neq 0$. Then there exists a region $R \subset (a, b)$ such that $p \in R$ and f with domain restricted to R is injective. Furthermore, $f^{-1}: f(R) \rightarrow R$ is differentiable at the point $f(p)$ and*

$$(f^{-1})'(f(p)) = \frac{1}{f'(p)}.$$

Proof of Theorem 12.20. We want to show that there exists a region $R \subset (a, b)$ with $p \in R$ such that f restricted to R is injective, that $f^{-1}: f(R) \rightarrow R$ is differentiable at $f(p)$, and that $(f^{-1})'(f(p)) = \frac{1}{f'(p)}$.

Since f' is continuous and $f'(p) \neq 0$, there exists a region $R \subset (a, b)$ with $p \in R$ such that $f'(x) \neq 0$ for all $x \in R$. Since f' is continuous on R and never zero, it cannot change sign, on R , and so f' is either strictly positive or strictly negative throughout. Then, considering **Remark 12.17**, we have that f is strictly monotone on R , and thereby injective on R simply by the definition of injectivity. Since $f|_R : R \rightarrow f(R)$ is bijective, $f^{-1} : f(R) \rightarrow R$ exists and is continuous.

Now, we want to show that f^{-1} is differentiable at $f(p)$. For this it suffices to show that

$$(f^{-1})'(f(p)) = \lim_{y \rightarrow f(p)} \frac{f^{-1}(y) - p}{y - f(p)}$$

exists.

Let $\varphi(x)$ be an auxiliary function defined as

$$\varphi(x) = \begin{cases} \frac{f(x) - f(p)}{x - p} & \text{if } x \neq p \\ f'(p) & \text{if } x = p \end{cases}$$

By definition of the function we have that φ is continuous at p , and since $f'(p) \neq 0$, we have $\varphi(p) \neq 0$.

Now, fix $y \in f(R)$. Then, let $x = f^{-1}(y)$. Then, we can rewrite this as $f(x) = y$, and thereby we get that $f(x) - f(p) = y - f(p)$.

Then, we can therefore rewrite the difference quotient as $\frac{x - p}{f(x) - f(p)} = \frac{1}{\varphi(x)}$. Furthermore, since f^{-1} is continuous, as $y \rightarrow f(p)$ we have $x = f^{-1}(y) \rightarrow f^{-1}(f(p)) = p$. Thus we can rewrite the limit to get

$$\lim_{y \rightarrow f(p)} \frac{f^{-1}(y) - p}{y - f(p)} = \lim_{x \rightarrow p} \frac{x - p}{f(x) - f(p)} = \lim_{x \rightarrow p} \frac{1}{\varphi(x)} = \frac{1}{\varphi(p)} = \frac{1}{f'(p)}$$

As we have shown the limit exists, f^{-1} is differentiable at $f(p)$ and $(f^{-1})'(f(p)) = \frac{1}{f'(p)}$.

This suffices to complete the proof. □

Hint: Remark 12.17 and Lemma 11.8 should be useful.

Exercise 12.21. Consider the function $f(x) = x^n$ for a fixed $n \in \mathbb{N}$. Show that if n is even, then f is strictly increasing on the set of non-negative real numbers and that if n is odd, then f is strictly increasing on all of $\mathbb{R}$. For a given n , let A be the aforementioned set on which f is strictly increasing. Define the inverse function $f^{-1} : f(A) \rightarrow A$ by $f^{-1}(x) = \sqrt[n]{x}$, which we

sometimes also denote $f^{-1}(x) = x^{1/n}$. Use Theorem 12.20 to find the points $y \in f(A)$ at which f^{-1} is differentiable, and determine $(f^{-1})'(y)$ at these points.

Proof of Exercise 12.21. We want to show that if n is even, then $f(x) = x^n$ is strictly increasing on $\{x \in \mathbb{R} : x \geq 0\}$, and if n is odd, then f is strictly increasing on $\mathbb{R}$.

Suppose $n \in \mathbb{N}$ is even. Consider $x_0, x_1 \geq 0$. Without loss of generality, say $x_0 < x_1$. For $n \in \mathbb{N}$, by definition, we have that $0 < x_0^n < x_1^n$, hence $f(x_0) < f(x_1)$. Therefore f is strictly increasing on the set of non-negative real numbers.

Now suppose $n \in \mathbb{N}$ is odd. Then $n = 2m + 1$ for some $m \in \mathbb{N} \cup \{0\}$. Consider $x_0, x_1 \in \mathbb{R}$ with $x_0 < x_1$. We consider three cases.

Case 1: $0 \leq x_0 < x_1$. This follows by the same argument as the even case, hence $f(x_0) < f(x_1)$.

Case 2: $x_0 < 0 < x_1$. We have that $x_0^n = x_0^{2m+1} = x_0^{2m} \cdot x_0$. Since $x_0 < 0$ and $x_0^{2m} > 0$, we get $x_0^n < 0$. Similarly, since $x_1 > 0$, we get $x_1^n > 0$. Hence $f(x_0) < 0 < f(x_1)$.

Case 3: $x_0 < x_1 < 0$. Since $x_0 < x_1 < 0$, we have $|x_0| > |x_1|$. Since both are non-negative, $|x_0|^n > |x_1|^n$. Since n is odd, $x_0^n = -|x_0|^n$ and $x_1^n = -|x_1|^n$, hence $x_0^n = -|x_0|^n < -|x_1|^n = x_1^n$, and therefore $f(x_0) < f(x_1)$.

Thus, for all $x_0, x_1 \in \mathbb{R}$ with $x_0 < x_1$, we have $f(x_0) < f(x_1)$, hence f is strictly increasing on $\mathbb{R}$.

Now, we apply Theorem 12.20 to find the points at which f^{-1} is differentiable. We define $f^{-1} : f(A) \rightarrow A$ by $f^{-1}(x) = \sqrt[n]{x} = x^{1/n}$. By Theorem 12.20, f^{-1} is differentiable at $f(a)$ for all $a \in A$ such that $f'(a) \neq 0$. We have that $f'(a) = na^{n-1}$, hence $f'(a) \neq 0$ whenever $a \neq 0$, and thereby $f(a) \neq 0$. Therefore f^{-1} is differentiable at all $y \in f(A)$ with $y \neq 0$, and

$$(f^{-1})'(y) = \frac{1}{f'(f^{-1}(y))} = \frac{1}{n(f^{-1}(y))^{n-1}} = \frac{1}{n(\sqrt[n]{y})^{n-1}},$$

This completes the proof. □

Additional Exercises

In these exercises you may assume familiarity with the trigonometric functions and the logarithmic/exponential functions and their derivatives.

1. Use Rolle's Theorem (12.14) to show that a polynomial of degree $n \geq 1$ has at most n distinct roots.

Additional Exercise 1. We want to show that a polynomial of degree $n \geq 1$ has at most n distinct roots. We want to show this through a proof by induction on n .

Let f be a polynomial in one variable. Firstly, we must consider the base case. Suppose $n = 1$. Then we can write f as $f(x) = ax + b$ for some $a \neq 0$. By definition, we have that $f(x)$ has exactly one root $x = -\frac{b}{a}$. Hence f has at most 1 root.

Now, assume that every polynomial of degree $n - 1$ has at most $n - 1$ distinct roots. Let f be a polynomial of degree n . Suppose for contradiction that f has at least $n + 1$ distinct roots $x_1 < x_2 < \dots < x_{n+1}$.

Since f is a polynomial, it is continuous and differentiable on $\mathbb{R}$. Consider the n consecutive pairs $(x_1, x_2), (x_2, x_3), \dots, (x_n, x_{n+1})$. For each consecutive pair (x_i, x_{i+1}) , we have $p(x_i) = p(x_{i+1}) = 0$ where $i \in [n]$. Then, by Rolle's Theorem, we have that there exists a point $c_i \in (x_i, x_{i+1})$ such that $f'(c_i) = 0$. Since the intervals $(x_1, x_2), (x_2, x_3), \dots, (x_n, x_{n+1})$ are all distinct, the points $c_1, c_2, \dots, c_n$ are distinct.

However, p' is a polynomial of degree $n - 1$, and by our inductive assumption, every polynomial of degree $n - 1$ has at most $n - 1$ distinct roots. Hence p' has at most $n - 1$ distinct roots. But we have shown that p' has at least n distinct roots, and since $n > n - 1$, this is a contradiction. Therefore our assumption that p has at least $n + 1$ distinct roots must be false, and so p has at most n distinct roots, which completes the proof. $\square$

2. Suppose that f is differentiable on an interval $[x_1, x_2]$ for $0 < x_1 < x_2$. Prove that there exists a point $t \in (x_1, x_2)$, such that

$$\frac{x_1 f(x_2) - x_2 f(x_1)}{x_1 - x_2} = f(t) - t f'(t).$$

3. (a) Suppose $f: [a, b] \rightarrow \mathbb{R}$, $x_0 \in (a, b)$, f', f'' exist and are continuous on some open interval containing x_0 , and $f'(x_0) = 0$, but $f''(x_0) \neq 0$. Prove that
- i) if $f''(x_0) > 0$ then f has a local minimum at x_0 ;
 - ii) if $f''(x_0) < 0$ then f has a local maximum at x_0 .
- (b) **Bernoulli's inequality** Prove that for $x \geq -1$ and $\alpha \geq 1$

$$(1 + x)^\alpha \geq 1 + \alpha x.$$

and for $x \geq -1$ and $0 \leq \alpha \leq 1$,

$$(1 + x)^\alpha \leq 1 + \alpha x.$$

- (c) Prove that $x - \frac{x^3}{6} < \sin(x) < x$, for $x > 0$.

Additional Exercise 3(a.) Suppose $f : [a, b] \rightarrow \mathbb{R}$, $x_0 \in (a, b)$, f' , f'' exists and are continuous on some open interval containing x_0 , and $f'(x_0) = 0$, but $f''(x_0) \neq 0$.

Firstly, we want to show that if $f''(x_0) > 0$ then f has a local minimum at x_0 . If $f''(x_0) > 0$ then, by **Corollary 12.16**, we have that f' is strictly increasing on $[a, b]$. Now, consider $x_0 \in (a, b)$. As f' is strictly increasing, we can say that $f'(a) < f'(x_0) = 0 < f'(b)$. Therefore, it must be that $f'(a) < 0$.

Now consider an arbitrary region R that contains x_0 . We can define this region R as $(x_0 - \delta, x_0 + \delta)$ where $\delta > 0$. Then, by definition $R \subset A$. Therefore, $f(A \cap R) = f(R)$. As f' is strictly increasing, we have that $f'(A)$ has a first point x_0 . Thus, we have that x_0 becomes the local minimum point.

Secondly, we want to show that if $f''(x_0) < 0$ then f has a local maximum at x_0 . If $f''(x_0) < 0$ then, by **Corollary 12.16**, we have that f' is strictly decreasing on $[a, b]$. Now, consider $x_0 \in (a, b)$. As f' is strictly decreasing, we can say that $f'(a) > f'(x_0) = 0 > f'(b)$. Therefore, it must be that $f'(b) < 0$.

Now consider an arbitrary region R that contains x_0 . We can define this region R as $(x_0 - \delta, x_0 + \delta)$ where $\delta > 0$. Then, by definition $R \subset A$. Therefore, $f(A \cap R) = f(R)$. As f' is strictly decreasing, we have that $f'(A)$ has a last point x_0 . Thus, we have that x_0 becomes the local maximum point.

□

Additional Exercise 3(b.) We want to prove that for $x \geq -1$ and $\alpha \geq 1$, $(1+x)^\alpha \geq 1 + \alpha x$. For this, let $g(x) = (1+x)^\alpha - 1 - \alpha x$. It suffices to show that $g(x)$ is strictly increasing.

To show that $g(x)$ is strictly increasing, by **Corollary 12.16**, it suffices to show that $g'(x) > 0$ for all $x \in \mathbb{R}$. By definition of $g(x)$, we have that it is differentiable and continuous across ... By **Corollary 12.15**, we have that $g'(\lambda)$, where $\lambda \in \mathbb{R}$ can be defined as $\frac{g(b)-g(a)}{b-a}$. As $g(x) = (1+x)^\alpha - 1 - \alpha x$, we can write this as

$$g'(\lambda) = \frac{((1+b)^\alpha - 1 - \alpha b) - ((1+a)^\alpha - 1 - \alpha a)}{b-a}$$

.

$$g'(\lambda) = \frac{(1+b)^\alpha - \alpha b + \alpha a}{b-a}$$

As, by definition of the interval $[a, b]$ and (a, b) , we have that $b > a$, therefore, we get $-\alpha b + \alpha a > -\alpha b$. Thus, for $g'(\lambda) > 0$, it suffices to show that $(1 + b)^\alpha > \alpha b$. $\square$

Additional Exercise 3(c.) We want to show that $x - \frac{x^3}{6} < \sin(x) < x$ for $x > 0$. For this, we will separate the inequality. Firstly we will prove that $\sin(x) < x$ and then secondly we will show that $x - \frac{x^3}{6} < \sin(x)$.

Firstly, We want to show that $\sin(x) < x$ for $x > 0$. For this we can define $f(x) = x - \sin(x)$. Then we get that $f(0) = 0 - \sin(0) = 0$. Now, consider, $f'(x)$. We have that $f'(x) = 1 - \cos(x) \geq 0$ for all x , and thereby $f'(x) > 0$ for all $x > 0$. Therefore, by **Corollary 12.16**, f is strictly increasing on $(0, \infty)$. Hence for all $x > 0$, we have $f(x) > f(0) = 0$, and therefore $x - \sin(x) > 0$, which gives us $\sin(x) < x$.

Secondly, we want to show that $x - \frac{x^3}{6} < \sin(x)$ for $x > 0$. We can define $g(x) = \sin(x) - x + \frac{x^3}{6}$. Then we get that $g(0) = \sin(0) - 0 + 0 = 0$. Now, we can consider $g'(x)$. We may write $g'(x)$ as:

$$g'(x) = \cos(x) - 1 + \frac{x^2}{2}.$$

By definition of the function we have that $g'(0) = \cos(0) - 1 + 0 = 0$. We can take the second derivative to get:

$$g''(x) = -\sin(x) + x.$$

From the first part, we have that $-\sin(x) + x > 0$ for all $x > 0$. Therefore, we have that $g''(x) > 0$ for all $x > 0$. Hence, by **Corollary 12.16**, g' is strictly increasing on $(0, \infty)$. Hence for all $x > 0$, we have $g'(x) > g'(0) = 0$. Therefore, g is strictly increasing on $(0, \infty)$. Hence for all $x > 0$, we have $g(x) > g(0) = 0$, and therefore $\sin(x) - x + \frac{x^3}{6} > 0$, which gives us $x - \frac{x^3}{6} < \sin(x)$.

This suffices to prove that $x - \frac{x^3}{6} < \sin(x) < x$ for all $x > 0$. $\square$

4. Let $f : [0, 1] \rightarrow [0, 1]$ be a continuous function.
 - i) Show that there is at least one $x \in [0, 1]$ such that $f(x) = x$.
 - ii) Show that if f is differentiable on $(0, 1)$ and $f'(x) \neq 1$, for all $x \in (0, 1)$, then there is a unique $x_0 \in [0, 1]$ such that $f(x_0) = x_0$.
5. (a) Prove that if f is differentiable on $[a, b]$ and $f'_+(a) < 0$, then there is a point $x \in (a, b)$ with $f(x) < f(a)$. Similarly, if $f'_-(b) > 0$, there is a point $y \in (a, b)$ with $f(y) < f(b)$.

- (b) Prove that if f is differentiable on $[a, b]$ and $f'_+(a) < 0 < f'_-(b)$, then there is a point $c \in (a, b)$ with $f'(c) = 0$.
- (c) Prove that if f is differentiable on $[a, b]$, then for any α between $f'_+(a)$ and $f'_-(b)$ there is some $c \in (a, b)$ such that $f'(c) = \alpha$.

Additional Exercise 5(a.) We want to show that if f is differentiable on $[a, b]$ and $f'_+(a) < 0$, then there is a point $x \in (a, b)$ with $f(x) < f(a)$.

Firstly, we have that $f'_+(a) < 0$. Therefore, we can say that

$$\lim_{h \rightarrow 0^+} \frac{f(a+h) - f(a)}{h} < 0.$$

By **Lemma 11.8**, there exists $\delta > 0$ such that $\frac{f(a+h)-f(a)}{h} < 0$ for all $h \in (0, \delta)$. Since $f'_+(a) = \lim_{h \rightarrow 0^+} \frac{f(a+h)-f(a)}{h} < 0$, we can apply **Lemma 11.8** to the function $\frac{f(a+h)-f(a)}{h}$, which guarantees that the difference quotient remains negative in an entire neighborhood of $h = 0$, and not just in the limit. Hence, there exists $\delta > 0$ such that $\frac{f(a+h)-f(a)}{h} < 0$ for all $h \in (0, \delta)$. Since $h > 0$, it follows that $f(a+h) - f(a) < 0$, and therefore $f(a+h) < f(a)$. Then we can fix $x = a + h$, so that $x \in (a, b)$ and $f(x) < f(a)$.

Secondly, we want to show that if $f'_-(b) > 0$, there is a point $y \in (a, b)$ with $f(y) < f(b)$. As $f'_-(b) > 0$, we have that

$$\lim_{h \rightarrow 0^-} \frac{f(b+h) - f(b)}{h} > 0.$$

By **Lemma 11.8**, we have that there exists $\delta > 0$ such that $\frac{f(b+h)-f(b)}{h} > 0$ for all $h \in (-\delta, 0)$. Since $h < 0$, it follows that $f(b+h) - f(b) < 0$, and therefore $f(b+h) < f(b)$. Then, we can fix $y = b + h$ for any such h and we have $y \in (a, b)$ and $f(y) < f(b)$. $\square$

Additional Exercise 5(b.) We want to show that if f is differentiable on $[a, b]$ and $f'_+(a) < 0 < f'_-(b)$, then there is a point $c \in (a, b)$ with $f'(c) = 0$.

By part (a), since $f'_+(a) < 0$, there exists a point $x \in (a, b)$ such that $f(x) < f(a)$. Similarly, since $f'_-(b) > 0$, there exists a point $y \in (a, b)$ such that $f(y) < f(b)$. Therefore f does not attain its minimum on $[a, b]$ at either endpoint. Since f is continuous on $[a, b]$, by the Extreme Value Theorem f attains its minimum at some point $c \in (a, b)$. Since c is in the interior, $f(c)$ is a local minimum, and therefore by Theorem 12.13, $f'(c) = 0$. $\square$

Additional Exercise 5(c.) We want to show that if f is differentiable on $[a, b]$, then for any α between $f'_+(a)$ and $f'_-(b)$ there is some $c \in (a, b)$ such that $f'(c) = \alpha$.

Define $g(x) = f(x) - \alpha x$. Then g is differentiable on $[a, b]$ and $g'(x) = f'(x) - \alpha$. We compute the one-sided derivatives at the endpoints. We have that

$$g'_+(a) = f'_+(a) - \alpha \quad \text{and} \quad g'_-(b) = f'_-(b) - \alpha.$$

Since α is between $f'_+(a)$ and $f'_-(b)$, we have that $g'_+(a) < 0 < g'_-(b)$. Therefore by part (b), there exists a point $c \in (a, b)$ such that $g'(c) = 0$, i.e. $f'(c) - \alpha = 0$, and therefore $f'(c) = \alpha$. $\square$

6. (a) Suppose that f and g are continuous on $[a, b]$, differentiable on (a, b) , that $f(a) = g(a) = 0$ and $g(x) \neq 0, g'(x) \neq 0$ for $x \in (a, b)$. Prove that

$$\text{if } \lim_{x \rightarrow a} \frac{f'(x)}{g'(x)} = L, \quad \text{then } \lim_{x \rightarrow a} \frac{f(x)}{g(x)} = L.$$

(b) Compute the following limits:

i)

$$\lim_{x \rightarrow 0} \frac{\tan x - x}{\sin x - x},$$

ii)

$$\lim_{x \rightarrow 0} \frac{\tan 3x}{\tan x},$$

iii)

$$\lim_{x \rightarrow 0} \frac{1 - \cos(x^2)}{x^2 \sin x^2},$$

iv)

$$\lim_{x \rightarrow 0} \frac{\cos(\sin x) - \cos x}{x^4},$$

v)

$$\lim_{x \rightarrow 0} \frac{\cos x - e^{-\frac{x^2}{2}}}{x^4}.$$

7. (a) Let $n \in \mathbb{N}$ and let $f: [a, b] \rightarrow \mathbb{R}$ be such that $f, f', f'', \dots, f^{(n)}$ are continuous on $[a, b]$ and $f^{(n+1)}$ exists on (a, b) . Suppose that $f(a) = f(b)$ and $f'(a) = f''(a) = \dots = f^{(n)}(a) = 0$. Prove that there is some $c \in (a, b)$ such that $f^{(n+1)}(c) = 0$.
- (b) Let $n \in \mathbb{N}$ and let $f: [a, b] \rightarrow \mathbb{R}$ be such that $f, f', f'', \dots, f^{(n)}$ are continuous on $[a, b]$ and $f^{(n+1)}$ exists on (a, b) . Prove that if $x_0 \in [a, b]$, then for any $x \in [a, b]$ there is some λ between x_0 and x such that

$$\begin{aligned} f(x) &= f(x_0) + (x - x_0)f'(x_0) + \frac{(x - x_0)^2}{2!}f''(x_0) + \dots + \frac{(x - x_0)^n}{n!}f^{(n)}(x_0) \\ &\quad + \frac{(x - x_0)^{n+1}}{(n+1)!}f^{(n+1)}(\lambda). \end{aligned}$$

- (c) i) Expand $\tan(x)$ up to the term x^5 at $x = 0$,
 ii) Expand $\sin(\sin x)$ up to the term x^3 at $x = 0$.
 iii) Expand e^x up to the term x^4 at $x = 0$.

8. Compute the derivatives of the following functions:

a) $y = x + x^x + x^{x^x}$,

b) $y = \arctan(\tan^2(x))$,

c) $y = \sqrt{x + \sqrt{x + \sqrt{x}}}$,

d) $y = \ln \frac{x^2 - 1}{x^2 + 1}$,

e) $y = \ln(x + \sqrt{x^2 + 1})$,

f) $y = \ln \sqrt{\frac{1 - \sin x}{1 + \sin x}}$,

g) $y = \arctan \left(\frac{\sin x + \cos x}{\sin x - \cos x} \right)$.